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Roger Decanio
The Sutter Law Firm, PLLC
1598 Kanawha Boulevard, East
Charleston, WV 25311

RE: Well Sampling, Seth/Prenter

Dear Mr. Decanio:

This letter concerns the potential impacts to drinking water wells in the Boone County, WV areas of Seth and Prenter. On October 29, 2008 I had the opportunity to visit the more of the homes in the subject area. As during previous sampling, I also took the opportunity to question the residents about the drinking water in their homes. While this visit was limited in scope, based on these interviews, my previous and continued sampling, and the sampling work of Ben Stout (Wheeling-Jesuit University), it appears that these wells have been impacted by nearby mining activities.

As during the August sampling event, interviews with residence indicated a deterioration of water quality over the last few years. All reported taste, appearance and odor changes over that period, as well as the smell of “rotten eggs”. In all instances I did smell hydrogen sulfide gas in the home and/or the water sample.

Water wells in the area have historically been the sole source of drinking, cooking and bathing water. Recently, bottled water has been used for drinking and cooking in some homes.

Samples were collected from either the well pressure tank or, if the pressure tank was not accessible, from the tap. All samples were collected prior to treatment (softeners, etc.) if applicable. Odor and color was noted in each sample. Samples were analyzed for inorganic contaminants only. Contaminant levels were above applicable drinking water standards in many instances, including those standards for iron, barium, arsenic, manganese and lead. The presence of lead and arsenic is particularly troubling, as the toxicity of these contaminants has lead USEPA to establish contaminant level goals of zero for both, indicating that there is no safe level of exposure.

Generally, sampling has included those inorganic constituents associated with coal mining as well as those previously detected, especially those detected at levels of concern. In short, as with any sampling event, constituents to be analyzed had to be limited. There are undoubtedly other contaminants in the aquifer that were not analyzed for, such as organic contaminants – the main purpose of this limited sampling event and site visit was to determine if a widespread impact to the aquifer is likely to exist and to determine some of the toxics to which residents were exposed. The sampling events and site visits do indicate that impacts from mining activities are present, and a threat to human health. Additionally, the presence of nearby surface and underground mining, as well as slurry impoundments and underground slurry injection, indicates the source of these contaminants. Constituents and analytical results can be found in accompanying reports.

In addition to the concern of contaminants in the water supply, hydrogen sulfide gas is being generated in the groundwater and is making its way into these homes. As previously explained, the gas is formed through the reduction of sulfate, present in the areas groundwater, indicative of the local mining activities. Naturally occurring sulfate and iron reducing bacteria reduce the sulfate to sulfide in their normal metabolic activities. This creates hydrogen sulfide, a very toxic and pungent gas. Additionally, the presence of sulfide indicates strongly reducing conditions in and around the wells. This will have a tremendous impact on the form and availability of the contaminants in the well by changing their redox states. This in turn will effect solubility, mobility and toxicity of these metals. As conditions change, for example when residents use bleach to temporarily reduce the sulfide odor, redox conditions will change again. These changing redox conditions will have a huge effect of the variability of contaminant concentrations in the well water, as well as the mobility and toxicity of those contaminants.

<u>Slurry Injection/Slurry Impoundments</u>

There is no better way to move contaminants from a solid, such as rock, into water as a dissolved or suspended constituent than to grind it up to a fine particle size and mix it with the water. This is what occurs when coal slurry is produced. In addition to the fines from waste rock and coal, man-made organic contaminants are also introduced into the slurry.

Once in a slurry, contaminants such as iron, manganese, arsenic, lead and sulfate dissolve from the parent material, either coal or waste rock, and dissolve or are suspended in the liquid water, and other liquids, which create the slurry. This process is controlled by various factors, including solubility, temperature and contact time, but all contaminants will absolutely move from the solid to the liquid, dissolved phase in varying degrees, and many will be suspended as fine solids in the solution. Additionally, natural waters coming in contact with the slurry will also be contaminated.

In this case, this slurry was disposed of via injection wells over several years, and may be continuing. Additionally, slurry impoundments are in use in the vicinity, and are certainly contributing to groundwater contaminantion. The injection wells deposited the slurry in abandoned underground mines in the subject area. Slurry injection as a waste containment tool relies on the slurry remaining in the underground workings where it is placed, but that can only occur if the mine walls, floors and roofs are perfectly intact. However, the Appalachian Plateau, especially in coal-bearing strata and including the Seth/Prenter area, is known for having high secondary permeability because of fractures, joints, beading planes etc. Mining further enhances these features, as well as contributing its own, especially cracks. In this case, underground mining has occurred in several seams, and surface mining is currently being conducted in the area. Secondary permeability will therefore be much higher than primary permeability throughout the subject area, and extend well below the lowest mine works. Therefore slurry, as well as the now-contaminated water, will migrate away from the mine works into which it was injected, into the surrounding strata and aquifers, and into nearby pumping wells. Undoubtedly groundwater has been negatively impacted as a result of these activities in the subject area.

Migration

The area used for this waste disposal has been heavily mined. Boreholes have been scattered about the region for slurry injection, but also for exploration of oil, gas and coal. Boreholes alone would hydraulically connect upper and lower strata in the area. However, the effect of mining on vertical hydraulic conductivity is even more pronounced.

Mining exaggerates existing cracks and fractures in strata as well as creates new ones, creating secondary porosity that is much greater than the primary porosity of the undisturbed, native formations. Post-mining, roof falls in the mines create subsidence cracks, which can extend vertically 100 to 300 feet or more. Cracks and fractures from underground mining would create a network of preferential pathways extending from well above the highest mined seams to well below the lowest mined seam seam, and would be especially dense between the mined areas. Connection between seams is a reasonable certainty.

Given just natural gradients in the area, slurry and contaminated water would move generally toward the subject communities. However, to compound this movement and to increase that gradient and flow rate, the injection process creates an injection well mound, forcing water and slurry away from the injection point and increasing the hydraulic gradient. In addition, as the community water supply in the area consists of many wells, a "wall" of withdrawal wells would "pull" contaminated water and slurry towards them, also increasing the hydraulic gradient towards the withdrawal wells. The effect of this was certainly greatest when injection is occurring; however, withdrawal has never ceased and has likely had a huge impact on hydraulic gradient in the region, in effect pulling slurry and contaminated water towards the drinking water wells in the subject communities.

Slurry, slurry water, and natural waters that have come in contact with the slurry have undoubtedly migrated from the inject points into the drinking water wells in the subject communities.

Use

Water wells in the area have historically been the sole source of drinking, cooking and bathing water. Recently, bottled water has been used for drinking and cooking in some homes, and water is supplied for pickup at a local community center.

Unknowns

Historically, organic contaminants were introduced into the coal slurry from the processing facilities as an aid in flocculation. These contaminants included diesel fuel and acrylamides, both highly toxic carcinogens. Additionally, coal itself introduces organic contaminants into the environment, including polynuclear aromatic hydrocarbons, many of which are known or suspected carcinogens. As it is clear that slurry has impacted the wells in the region, it is also with certainty that in the residents have been exposed to these organic contaminants. To date, sampling for these contaminants has not been completed.

Based on this continuing review, I believe that coal mining and related activities has likely contaminated the drinking water supply in the Seth/Prenter area. This contaminated water migrated to the water supply wells in the subject communities and was used for consumption, bathing and cooking. This exposure has likely put these residents at increased risk of negative health effects.

As stated, work is ongoing, and subsequent reports will be provided. Should you have any questions, please feel free to contact me.

Sincerely;

D. Scott Simonton, PE, PhD